PROBLEM SET 1

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 40)

1) **(20 pts)** Given Ω and Λ are Hermitian, determine whether the following combinations are Hermitian, Anti-Hermitian or NOT Hermitian:

(a) **(5 pts)** $\Omega\Lambda$

(b) **(5 pts)** $\Omega\Lambda + \Lambda\Omega$

(c) **(5 pts)** $[\Omega, \Lambda]$

(d) **(5 pts)** $i[\Omega, \Lambda]$

2) **(5 pts)** Show that the product of two unitary operators, $(U_1,\ U_2)$, is unitary.

3) **(5 pts)** For a matrix $\Omega = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, where $\det\ \Omega = \alpha\delta - \beta\gamma$

(a) show that $\det\ \Omega^T = \det\ \Omega$

(b) show that $\det\ \Omega^T \Omega = \det\ \Omega^T \det\ \Omega$

(c) prove that the determinant of a unitary matrix, U, is a complex number of unit modulus, i.e. $|\det\ U| = 1$
hint: start with $\det\ (U^\dagger U)$ and use the properties shown in (a) and (b)

4) **(5 pts)** Verify that the operator, $R\left(\frac{\pi}{2}\mathbf{i}\right)$, is unitary (orthogonal) by examining its matrix.

5) **(5 pts)** Verify that the following matrices are unitary, and determine whether they are Hermitian or not

$$\frac{1}{\sqrt{2}}\begin{bmatrix}1 & i\\ i & 1\end{bmatrix}, \qquad \frac{1}{2}\begin{bmatrix}1+i & 1-i\\ 1-i & 1+i\end{bmatrix}$$